

Intelligent Agents Assignment

AI Study Preparation & Paper Prediction Agent

By the end of a semester, students often have more material than they can realistically review in equal depth. This assignment proposes an intelligent agent that works much like a careful teaching assistant: it reviews the student's own course material, notices topics and question styles that appear repeatedly, and turns those patterns into a clear, evidence-based order for revision.

1. Identify the Problem

Exam preparation often becomes a cycle of reading everything and hoping the most important ideas are remembered. Lecture notes, study guides, and past papers frequently revisit the same concepts, but identifying those patterns manually takes time. This can lead to uneven preparation: some topics receive too much attention, while other important areas are missed.

Problem statement: An intelligent agent is needed to examine a student's study material and past papers, identify recurring topics and question types, and convert that evidence into a practical, prioritized study plan. The agent should clearly present its output as an informed estimate rather than a guarantee of the exact exam paper.

2. Formulate the Problem

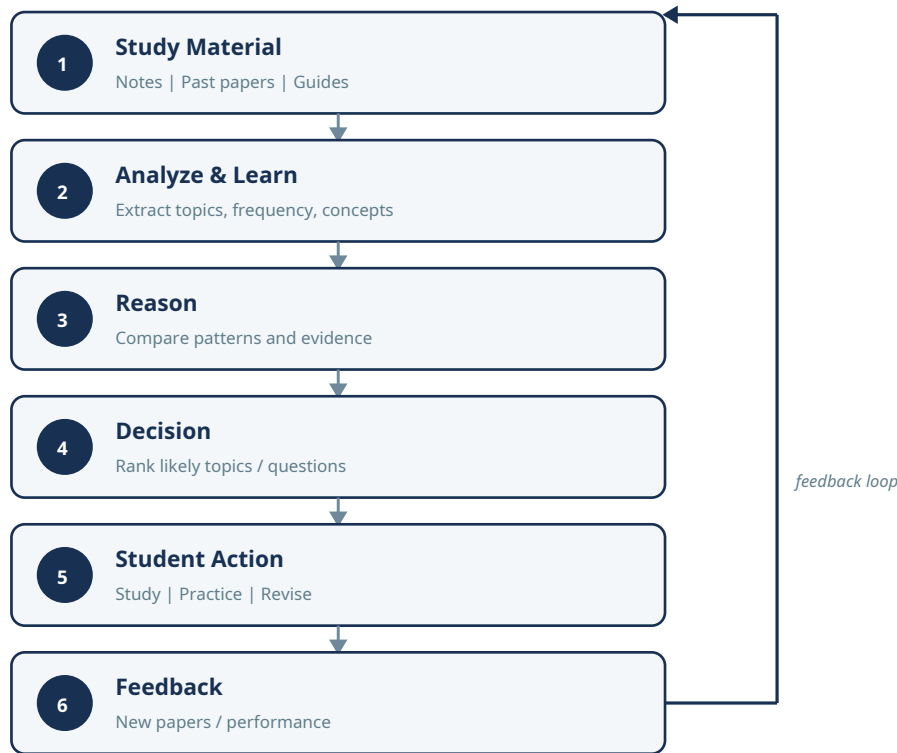
The problem can be described in terms of what the agent receives, what actions it can perform, the outcome it is trying to achieve, and the information it uses while making decisions.

Element	Description
Initial State	The agent begins with the material supplied by the student - study guides, lecture notes, past papers, and the course syllabus - without any prior assumption about which topics are most important.
Actions Available	Read and organize the material; extract topics and question types; compare past papers for recurring patterns; rank topics by importance; and generate a study plan with practice questions.
Goal State	The student receives a clear, ranked plan that explains what to study first and why, allowing preparation time to be focused on the areas most likely to matter.
Percepts	The text of notes and papers, topic and question labels, frequency of recurrence, recency, and feedback from quizzes or the student's own corrections.
Path of Actions	Collect material -> extract topics -> compare with past papers -> rank by evidence -> produce a study plan -> update when new material or results appear -> repeat.

In one sentence: Given a student's study guides, notes, and past papers, the agent should identify the topics that matter most and produce a practical revision plan - without claiming to know the exact exam in advance.

2.1 Graphical Problem Formulation

The workflow below shows the full cycle: study material enters the system, the agent analyzes and reasons over it, a ranked decision is produced, the student acts on that guidance, and the results return as feedback for the next cycle.



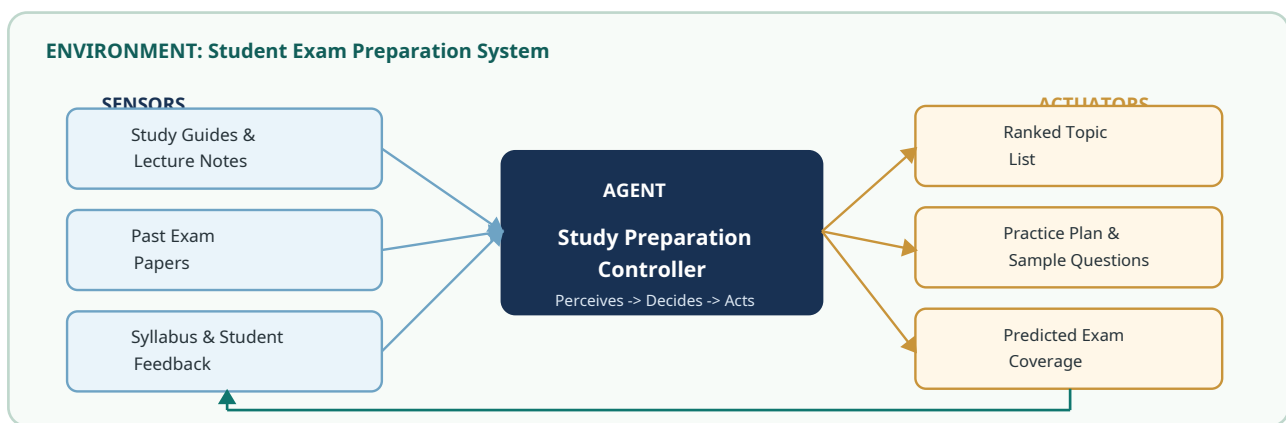
Learning -> Reasoning -> Decision Making (continuous feedback)

Study Material -> Analyze & Learn -> Reason -> Decision -> Student Action -> Feedback

2.2 Graphical Environment Representation

AI Study Preparation Agent - Perception-Action Cycle

This view places the agent inside its working environment. Sensors provide study guides, past papers, the syllabus, and student feedback. The controller interprets that information and produces useful outputs through its actuators: a ranked topic list, a practice plan with sample questions, and an estimate of likely exam coverage. As new material or performance feedback appears, it becomes fresh input for the next cycle.

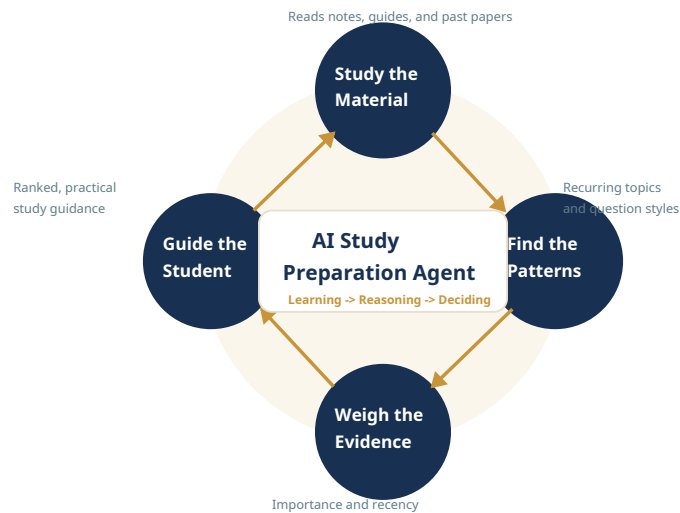


Student actions and new material create new percepts (feedback loop)

AI Study Preparation Agent - Environment, Sensors, and Actuators

2.3 The Agent's Cycle, at a Glance

At a high level, the process is simple and repetitive: review the material, detect recurring patterns, weigh the evidence, guide the student, and then repeat the cycle whenever new information becomes available.



3. Evaluate Using PEAS

PEAS provides a structured way to describe an intelligent agent by defining how it is evaluated, the environment in which it operates, how it acts, and what it senses.

Component	Applied to This Agent
Performance Measure	Accuracy and usefulness of topic rankings, coverage of important material, time saved during preparation, and whether exam performance improves.
Environment	The course environment: syllabus, lecture notes, study guides, and the history of past papers.
Actuators	A ranked priority list, a suggested study and practice schedule, and sample practice questions.
Sensors	Uploaded notes and past papers, the topics and question patterns found within them, and feedback from quizzes or student corrections.

4. Environmental Properties

The agent's environment can be classified using the standard properties below.

Property	Type	Why
Observability	Partially Observable	The agent only knows what appears in the material it receives. It cannot see the actual exam or know the examiner's private intentions.
Determinism	Stochastic	Past patterns can increase the likelihood of a topic appearing again, but they never guarantee it. New or unexpected content may still be introduced.
Episodes	Sequential	Today's study plan affects what the student knows tomorrow, so each recommendation builds on earlier actions rather than standing alone.
Change	Dynamic	New notes, past papers, and quiz results can appear at any time, causing the knowledge base to change independently of the agent's current action.
Values	Discrete & Continuous	Topics and question types are discrete categories, while confidence scores and frequency-based evidence can be represented continuously.
Agents	Single Agent	One intelligent agent supports the student; there is no separate agent competing or cooperating within the defined task.

5. What the Student Actually Gets

The final output is a straightforward three-level priority list - High, Medium, and Low - supported by the evidence behind each ranking and accompanied by a practical study plan. The agent should remain careful with its wording: a topic described as "likely to appear" is only a reasoned estimate based on observed patterns, not a promise about the exact paper.

HIGH PRIORITY Revise first	MEDIUM PRIORITY Revise next	LOW PRIORITY Review if time allows
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End of Assignment - AI Study Preparation & Paper Prediction Agent